

SCRAP

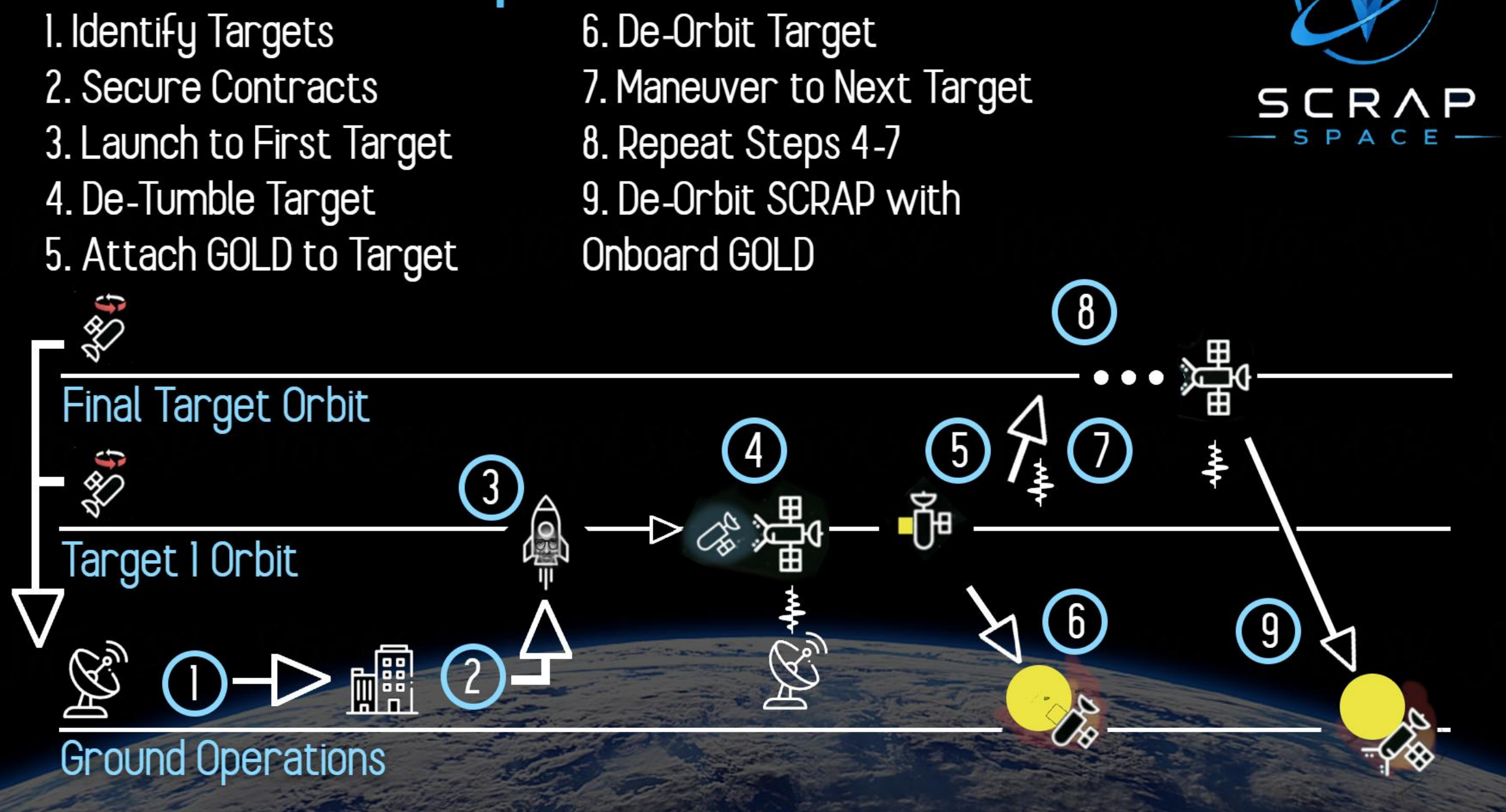
Satellite Control and Retrograde Acceleration Program

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Mission Overview

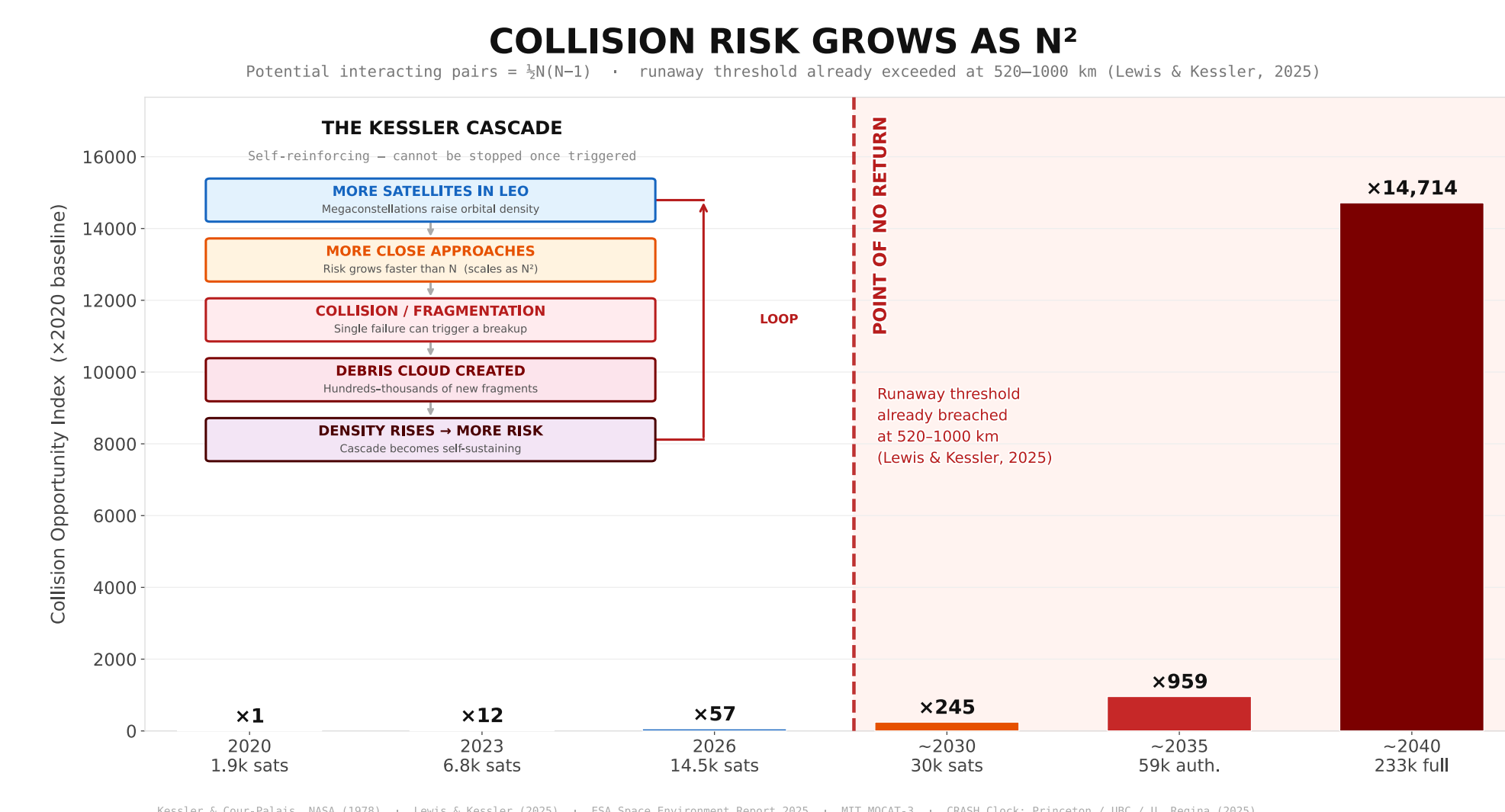
- Uncontrolled debris** in Low Earth Orbit (LEO) is a **serious and growing risk** to other satellites, and if left unmaintained could cascade beyond repair.
- SCRAP will **clear debris from LEO**, making space safer and more sustainable.
- SCRAP will autonomously rendezvous with debris, **de-tumble** it with an electromagnet, making it possible to attach an inflatable balloon which increases drag, causing **de-orbit**.

Mission ConOps



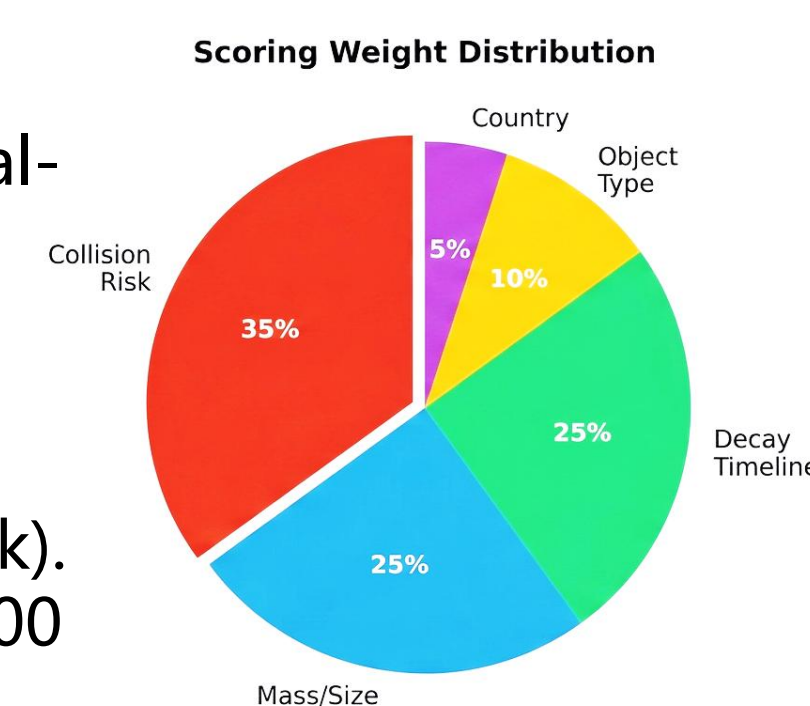
Business Case & Impact

- With enough debris accumulation, one collision could lead to an **irreversible and compounding cascading collision**, which could destroy every satellite within an orbit. This is called **Kessler Syndrome**.
- Kessler confirms satellite density between 520-1000km is already in a critical state. A catastrophic **collision cascade** would occur **today** without active maneuvering.
- Regulatory bodies are beginning to crack down on LEO operators; **FCC announced repercussions on operators** that cannot deorbit their satellites within **5 years** of mission complete.
- Private and Sovereign entities have provided **billions in funding** for de-orbiting solutions (ESA, Starfish Space, etc.).
- SCRAP is unique in that it can interact with debris via **non-contact methods**, particularly **contactless detumbling** which drastically reduces the risk of producing more debris.



Debris Selection

- SCRAP uses a custom catalog that combines satellite tracking and physical-property data to **rank LEO objects by removal value**.
- Custom pipeline **cross-references existing spacecraft data bases** for candidate sorting (DISCOS, Space-Track).
- Only considers objects between 400-700 km altitude.

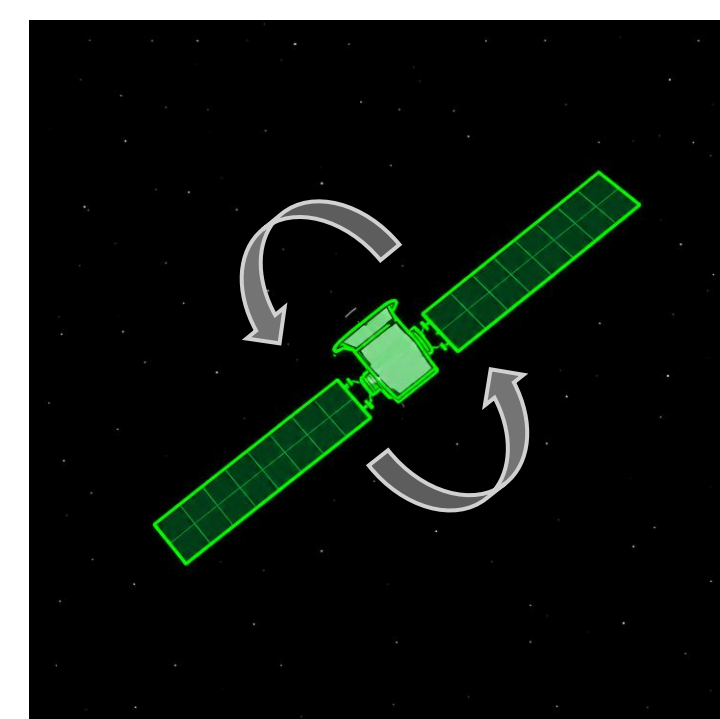


Guidance, Navigation, & Control

- Sensors** such as an IMU, star tracker, GPS receiver, time-of-flight sensor, and multiple cameras **inform location and orientation** with respect to both the Earth and the target debris.
- Reaction wheels** are used to **control orientation** and were designed to handle the momentum transfer associated with de-tumbling. **Monopropellant thrusters** allow for small corrective maneuvers. A **Hall-Effect Thruster** is used for **orbital transfers** and **station keeping**.

Debris Rotation Determination

- SCRAP uses a **vision-based optical flow pipeline** to characterize the target state.
- This process has 3 main steps:
 - Target segmentation
 - Feature tracking
 - Angular velocity estimation
- The debris rotation state is critical feedback for the control system that points the electromagnet at the target.



Mission Operations

- Operating a satellite in **close proximity** to another requires careful consideration of potential failures or collisions. The mission operations protocols have been designed to avoid faults and manage them if they occur.
- Redundant components** ensure measurement accuracy and backup options if a component fails.
- Satellite **communication architecture** prioritizes **access to mission status** for operators during proximity operations using a turnstile antenna.
- High data rate downlinks like video are planned outside of proximity operations using a parabolic dish.
- A "Go/No-Go" checklist has been defined for each critical mission stage based on health and target data downlinked from SCRAP.

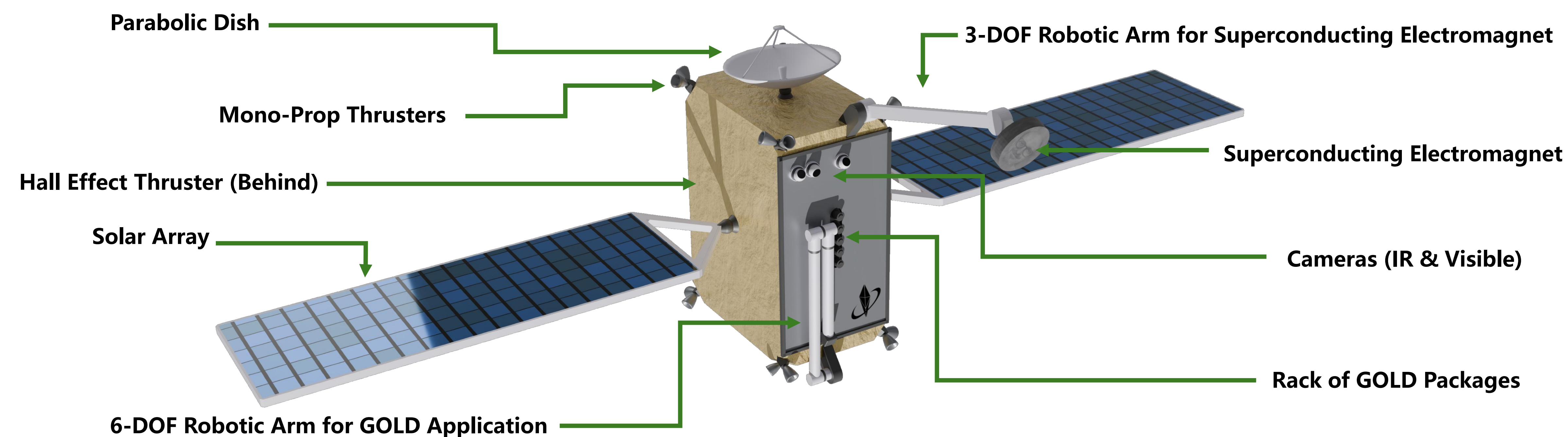
Risk Management

Rank	Risk ID	Approach	Risk Title	Criticality
1	R-001	M	Loss of Superconductivity During De-Tumbling Operations	High
2	R-002	M	Inability to Reduce Target Tumble Rate Within Required Time	High
3	R-003	M	Magnetic Interference with Sensitive Bus Components	Medium
4	R-006	M	GOLD Balloon Rupture and Collapse from debris	Medium
5	R-005	R	Incompatible Target Surface for Reliable GOLD Attachment	Medium
6	R-004	M	GOLD Package Detachment During Orbit-Lowering	Low

High Criticality Mitigation Strategies

- R-001: **Radiation shielding** to protect from solar heating and limits on current and B field density **reduce the risk of YBCO quenching**.
- R-002: In-situ entity observation allows **real-time candidate filtering** based on mission duration criteria and chaser capabilities to **prevent inefficient resource utilization**.

Satellite Hardware Breakdown



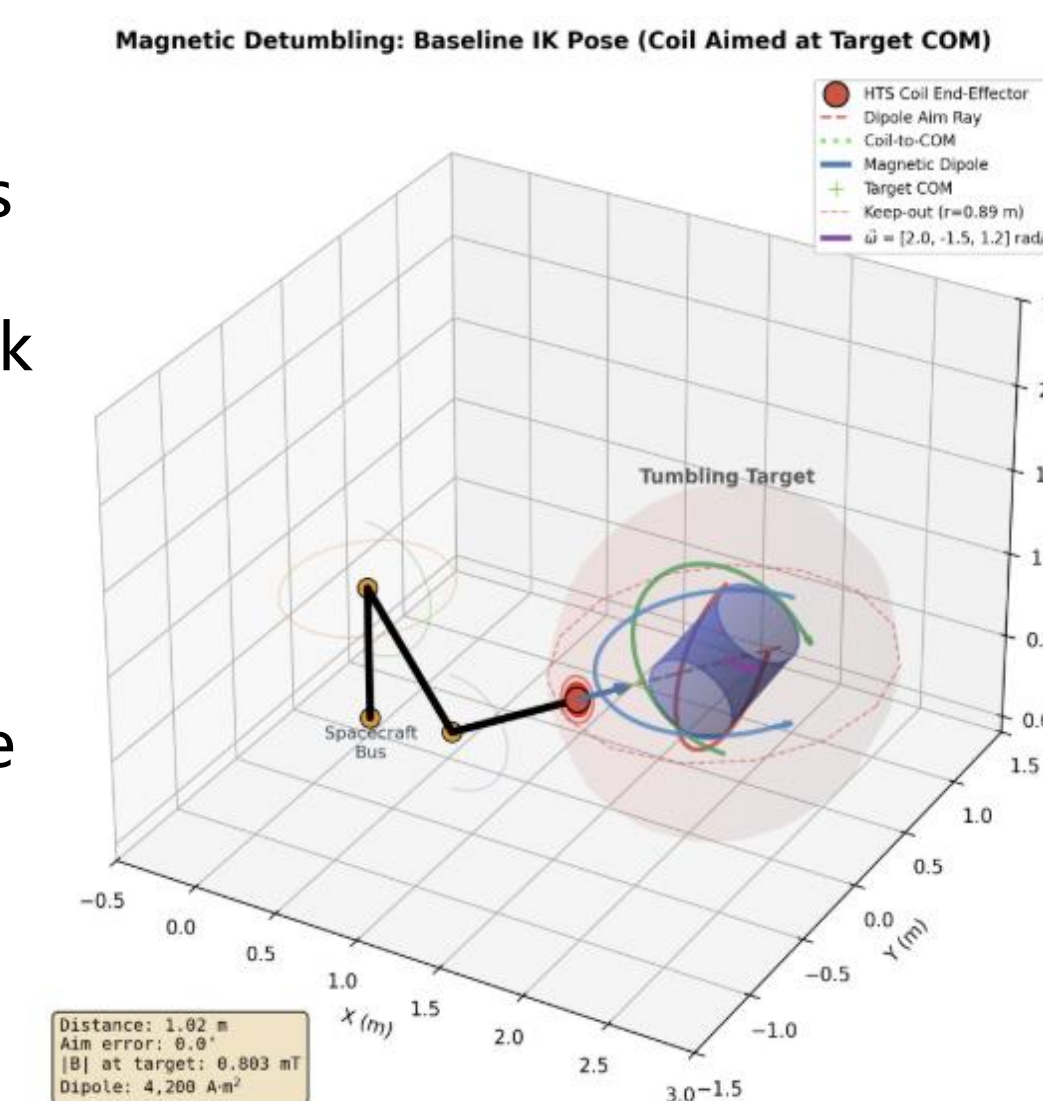
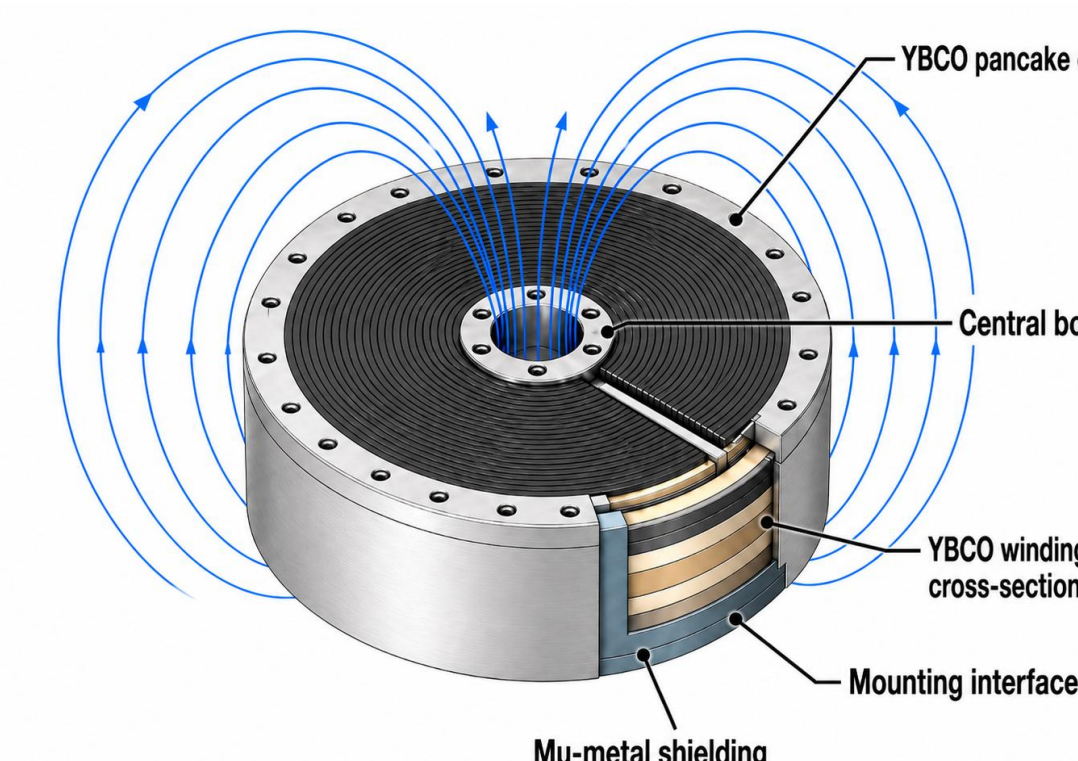
De-Tumble Solution: Electromagnetic Brake

De-Tumble Motivation

- Target rotation (tumble) makes interfacing dangerous; de-tumbling is therefore crucial.
- Non-contact braking** reduces our risk of creating excess debris.

Superconducting Electromagnet

- Eddy currents are generated by magnetic fields in rotating conductive metals, applying a counter rotation torque.
- A superconducting electromagnet is pointed perpendicular to the angular rotation vector for optimal torque.
- No propellant needed**; momentum is transferred to reaction wheels and desaturated by magnetorquers.
- Braking strength depends on distance, field strength, and effective magnetic tensor.

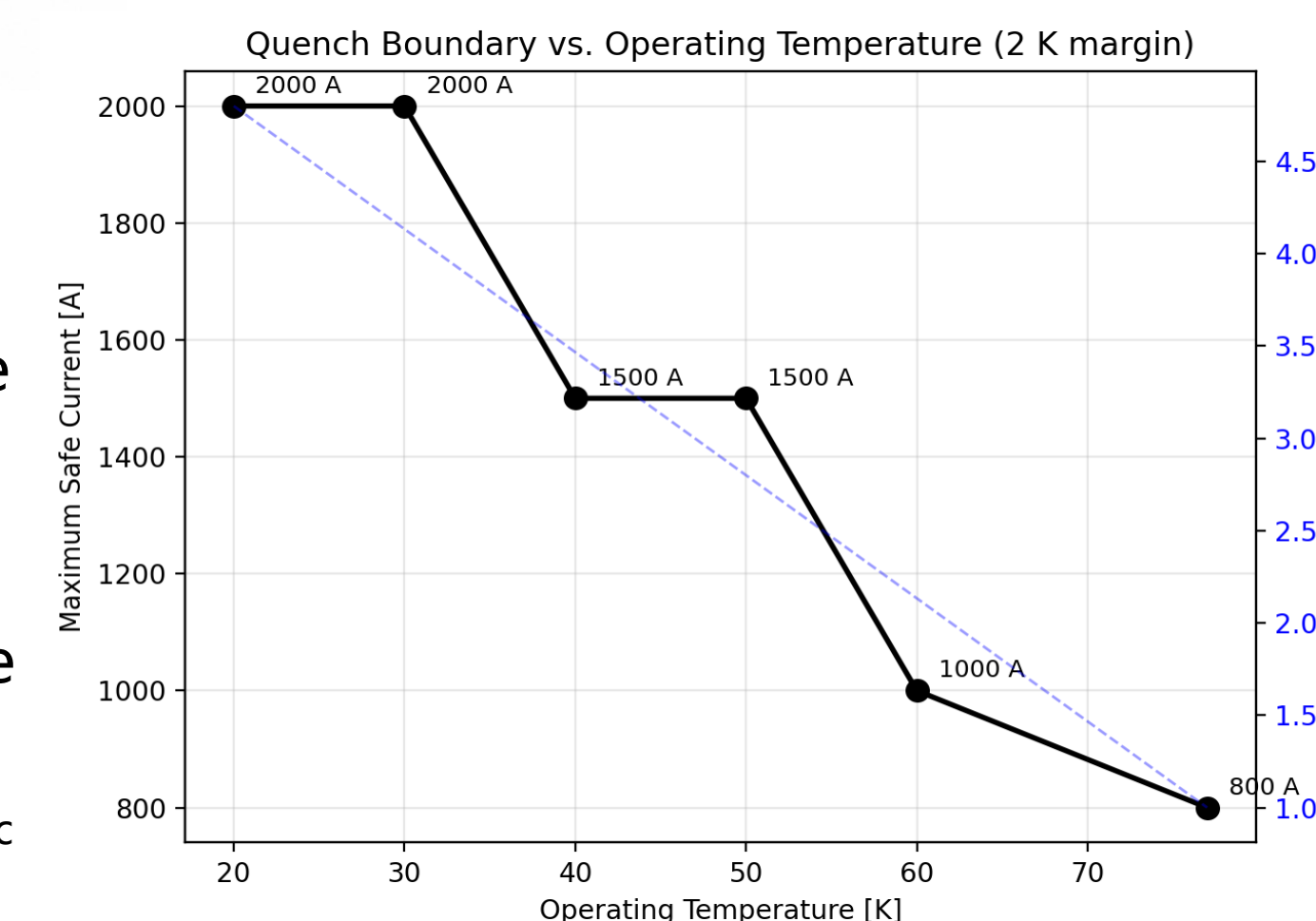


Coil Geometry & Design

- The geometry of the coil is a **20-turn YBCO pancake coil** meant to maximize the field strength in the direction perpendicular to the flat coil plane.
- The approximate magnetic dipole moment is 4200 Am²
- The approximate time to de-tumble using optimal criteria is on the **order of hours**.
- Mu metal shielding prevents excessive EMI to our Satellite.

Performance & Limitations

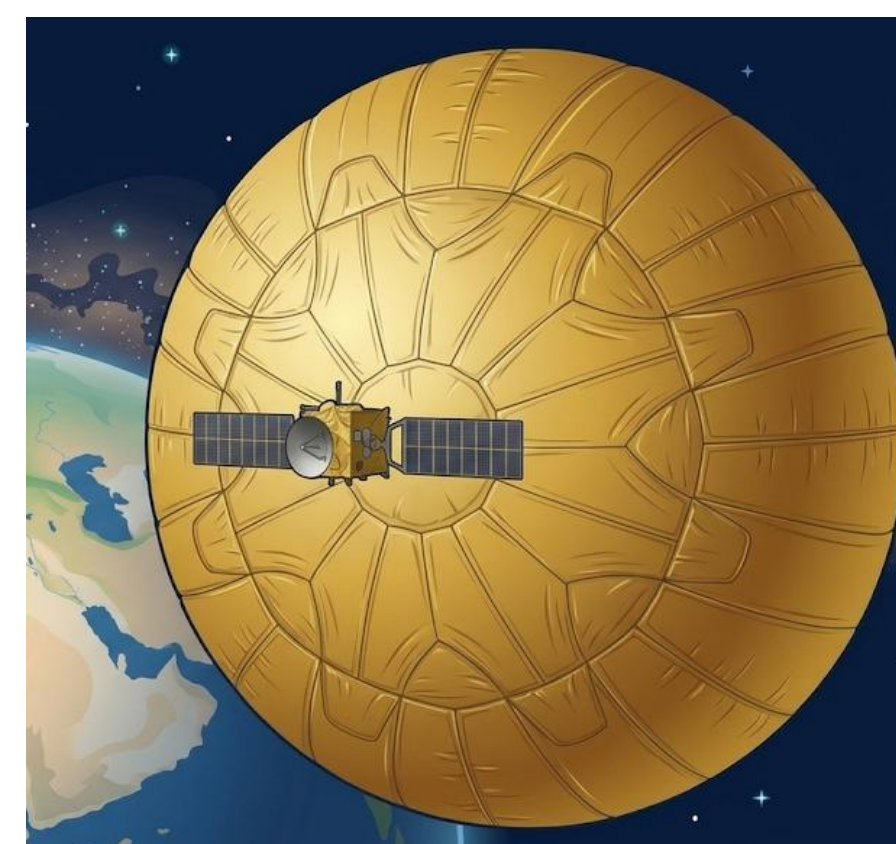
- The main risk of de-tumble failure is the **coil quenching**.
- Analytical sweep models were used in conjunction with a Finite Element Analysis verification model.
- Operating at **space-ambient temperatures** ($T < T_c$) allows the required B-field to be maintained without exceeding J_c or H_c , ensuring the system remains within the superconducting envelope and **preventing quenches**.



De-Orbit Solution: GOLD Package

Gossamer Orbit Lowering Device (GOLD)

- Passively de-orbits** debris with drag balloon technology.
- Inflatable reinforced Kapton balloon** has reference area determined from an atmospheric model calculating passive drag forces to de-orbit the candidate **within mission specific timeline**.

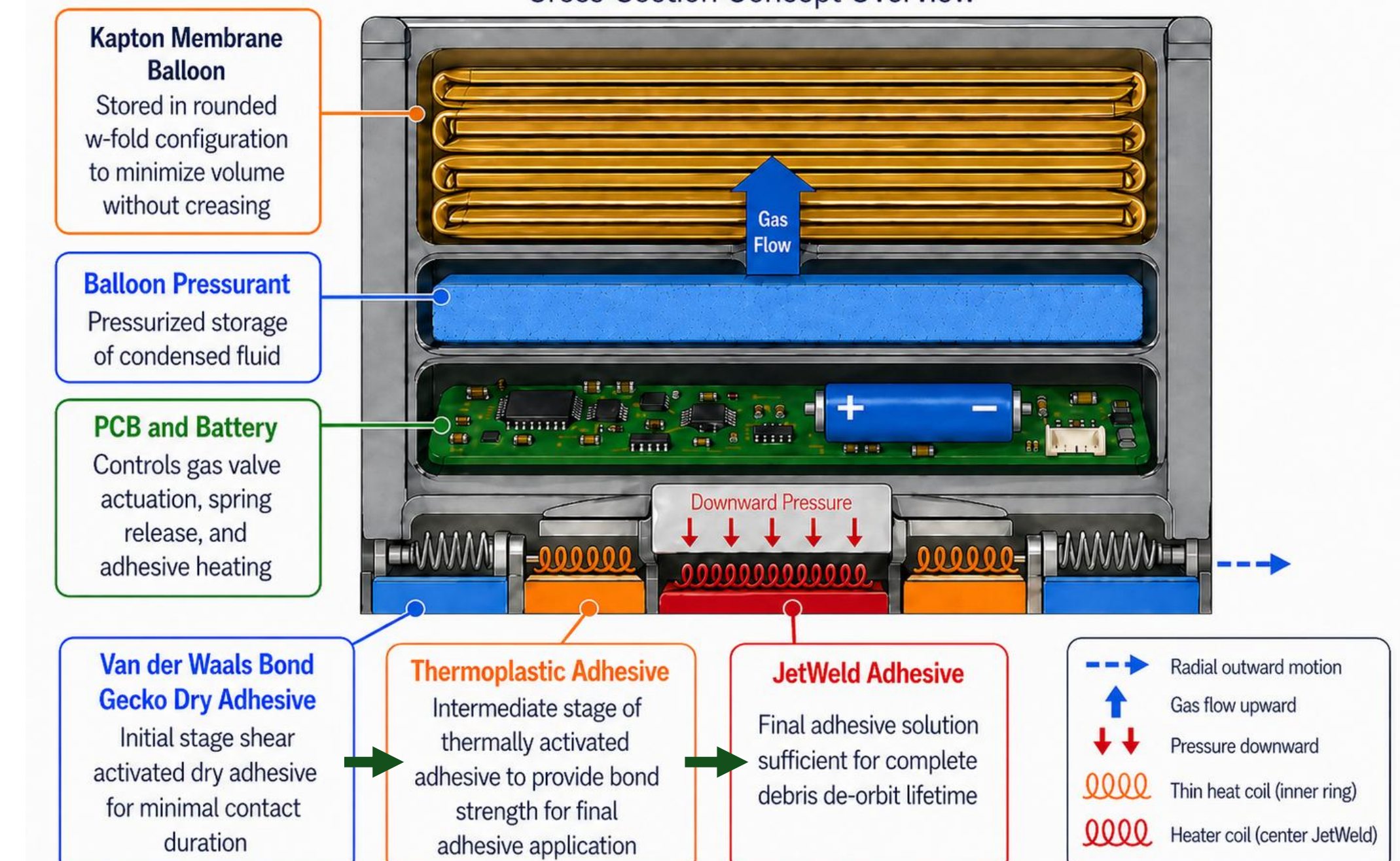
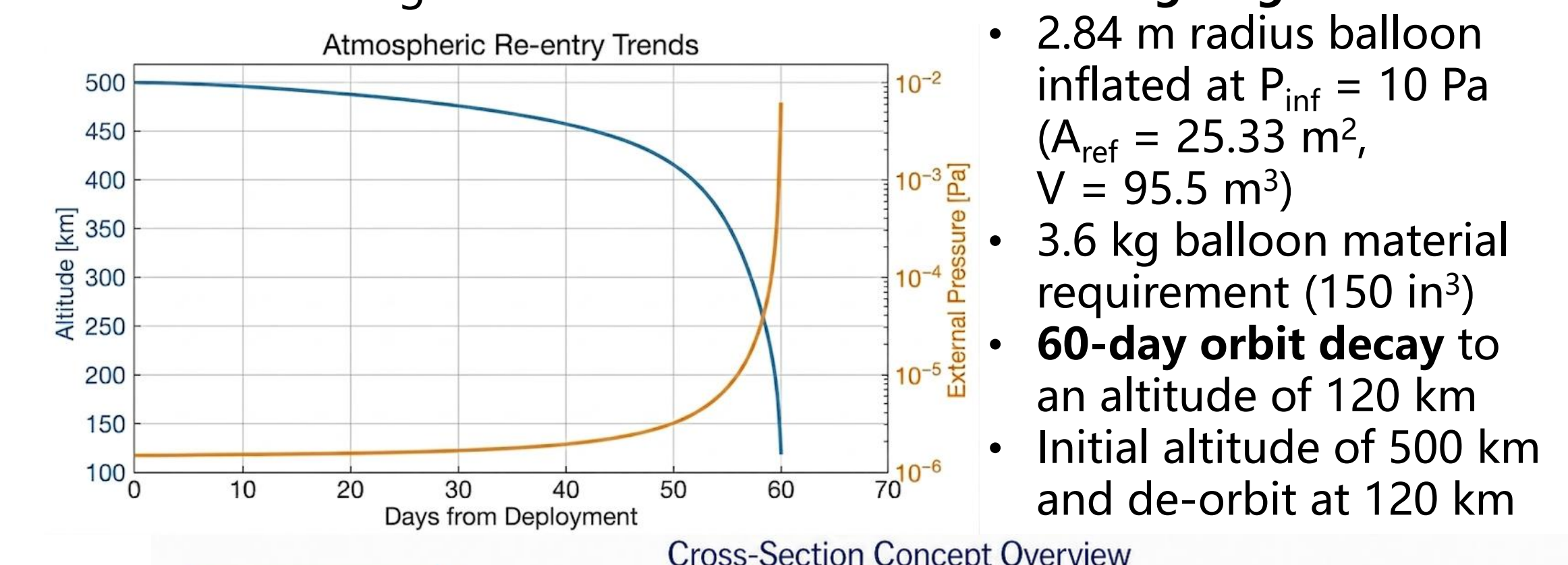


Attachment Method

- The GOLD balloon produces drag, requiring strong adhesion to the target.
- Staged adhesive** is necessitated by inertial nature of space environment to minimize momentum transfer between chaser and target.

Sized Reentry Trends

- 100 kg target** satellite
- 2.84 m radius balloon inflated at $P_{inf} = 10$ Pa ($A_{ref} = 25.33$ m², $V = 95.5$ m³)
- 3.6 kg balloon material requirement (150 in³)
- 60-day orbit decay** to an altitude of 120 km
- Initial altitude of 500 km and de-orbit at 120 km



Progression of Rendezvous Operations

